

Vikrant Tomar

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Research Profile

I am a PhD candidate in Atmospheric Sciences (thesis submission expected September 2026) with extensive hands-on experience in the operation, calibration, and maintenance of trace-gas instrumentation at a high-altitude atmospheric observatory in the Central Himalayas: gas analyzers (ozone, NO_x, NO_y, CO, SO₂), a thermal desorption GC-FID system for non-methane hydrocarbons, balloon-borne ozonesondes, and remote-sensing spectrometers (Pandora, MAX-DOAS, FTIR). I have planned and executed field campaigns for trace gas measurements and air sample collection across the Central Himalayas and the Indo-Gangetic Plain.

My research integrates in-situ measurements with satellite retrievals, reanalysis products, trajectory analysis, and atmospheric chemistry modelling to quantify ozone variability, precursor sensitivity and chemical regimes, and long-term trends across South Asia, supported by advanced data processing and statistical analysis in Python and R. I have experienced in mentoring students, science outreach, and publishing results through peer-reviewed publications and international conferences, with strong motivation to extend analytical expertise to mass-spectrometric measurements of gas and aerosol composition.

Research Interests

Atmospheric chemistry; tropospheric ozone and its precursors; air quality and air pollution; Atmospheric measurements and campaigns; remote sensing of air pollutants; regional transport and pathways.

Education

- **Ph.D. Candidate in Atmospheric Sciences**, Aryabhata Research Institute of Observational Sciences (ARIES); registered with Mahatma Jyotiba Phule Rohilkhand University. Aug 2021 – present; thesis submission expected September 2026. Uttarakhand, India.
- **M.Sc. in Physics and Astrophysics**, University of Delhi, Aug 2017-Jun 2019, New Delhi, India.
- **B.Sc. in Physics**, Ramjas College, University of Delhi, Aug 2014-Jun 2017, New Delhi, India.

Research Experience

Ph.D. Research: Spatial and temporal variation in tropospheric ozone over South Asia

- Investigated long-term (2007-2024) surface ozone variability and trends at a high-altitude site in the Central Himalayas using ground and satellite-based measurements and compared the observations with the regional model and global reanalysis output.
- Investigated the role of transport air masses in influencing surface ozone over the high altitude site in the central Himalayas. Transport models are combined with different techniques to identify transport pathways and source regions.
- Investigated the spatial and temporal variability in trace gases (Ozone, NO_x, NO_y, CO, NHMCs) across the Central Himalayas and Indo-Gangetic Plain using ground-based observations. Assessing the influence of precursor emissions, biomass burning, and regional transport at different sites.
- Investigated the impact of urbanization and development on tropospheric ozone and its precursor variability and trends across major Indian urban regions using satellite retrievals and reanalysis datasets over the last 2 decades.
- Investigates spatial and temporal variation in ozone and particulate matter and their chemical regimes over megacity Delhi through a framework integrating ground-based observations of ozone and PM_{2.5} with satellite-derived columnar levels of NO₂, HCHO, and NH₃.

Instrumentation and Laboratory Operations

- **Ground-based measurements:** Gas analyzers for Ozone, NO_x, NO_y, CO, and SO₂; Small alphasense sensors for PM_{2.5}, Ozone, NO_x, CO
- **Remote sensing measurement:** Pandora and MAX-DOAS spectroradiometers for columnar retrievals of Ozone, NO₂, and HCHO. FTIR spectrometer for column retrievals of greenhouse gases (CO₂, CH₄, etc.)
- **Balloon-borne measurements:** Ozone sonde and radiosonde for vertical profiling
- **Gas chromatography:** Operation of a thermal desorption-GC-FID system for hourly online NMHC measurements and offline analysis of campaign air samples
- **Calibration, maintenance, and QA/QC:** Responsibility for zero/span checks, multipoint calibrations using certified gas standards, flow calibration, leak checks, and troubleshooting of ozone, NO_x, NO_y, CO, and SO₂ analyzers.
- **Field campaigns:** Planned and executed trace-gas measurement campaigns and air-sample collections across the Central Himalayas and Indo-Gangetic Plain, including instrument deployment, site setup and logistics, and balloon-borne ozonesonde launches.

Technical Skills and Data Tools

Area	Experience
Programming	Python, R, Matlab, C, C++, Fortran
Operating systems	Linux, macOS, Windows, HPC environments
Data handling	SQL, Gnuplot, MS Excel; time-series analysis and scientific visualization
Instrumentation and QA/QC	Gas analyzer calibration (zero/span, gas standards, flow checks), TD-GC-FID operation, instrument troubleshooting
Transport and chemistry	HYSPLIT trajectory model, NCAR Master Mechanism
Satellite products	AIRS, OMI, OMI/MLS, TROPOMI, GEMS, MODIS fire radiative power and fire counts
Reanalysis and inventories	NCEP/GDAS, ERA5, CAMS, MERRA-2; EDGAR and CAMS emission inventories

Publications

Peer-Reviewed Publications

- Rajwar, M. C., Naja, M., Rawat, P., **Tomar, V.**, Kumar, M., Srivastava, P., & Lal, S. (2026). Non-methane hydrocarbon (NMHC) variability in rural Himalayan atmospheres: impacts on ozone, secondary organic aerosols, and human health. Environmental Science and Pollution Research, 1-14. <https://doi.org/10.1007/s11356-026-37694-z>
- Rajwar, M. C., Naja, M., Kant, Y., Rawat, P., **Tomar, V.**, Tiwari, R. K., & Lal, S. (2024). Ozone variability, its formation potential and crop losses in the Himalayan foothills. Air Quality, Atmosphere & Health, 1-14. <https://doi.org/10.1007/s11869-024-01572-9>

Conference Proceedings

- **Tomar, V.**, Naja, M., Lal, S., Rawat, P., & Kumar, U. (2024). Long-term changes in air pollutants at urban regions of India using remote sensing data. Proceedings of URSI-RCRS 2024. https://doi.org/10.46620/URSI_RSRC24/0994IDU7707
- Rajwar, M. C., Naja, M., Kant, Y., Rawat, P., **Tomar, V.**, Srivastava, P., & Lal, S. (2024). Unraveling ozone formation regimes in the Himalayan and IGP regions: A satellite-based study. Proceedings of URSI-RCRS 2024. https://doi.org/10.46620/URSI_RSRC24/0998APM8227

Submitted or Under Review

- **Tomar, V.**, Naja, M., Lal, S., Rawat, P., & Kumar, U. Long-term trends in air pollutants over major urban cities of India. (*Under review, Atmospheric Environment*)
- **Tomar, V.**, Naja, M., Kumar, M., Rawat, P., Lal, S., & Kumar, U. Long-term changes in ozone (2007-2024) at a high-altitude site in the Central Himalayas. (*Under review, Elementa*)
- **Tomar, V.**, Naja, M., Rawat, P., & Kumar, U. Spatiotemporal Analysis of Ozone and PM_{2.5} and Chemical Regimes over Delhi using Multi-Satellite and Surface Observations. (*Under review, Urban Climate*)

Manuscripts in Preparation

- Tomar, V., Naja, M., Kumar, M., Rawat, P., Lal, S., & Kumar, U. Influence of regional transport on surface ozone in the Central Himalayas. *In preparation.*
- Tomar, V., Naja, M., Kumar, M., Rawat, P., Rajwar, M., Lal, S., & Kumar, U. Surface ozone variability over the Indo-Gangetic Plain and Central Himalayas: Role of precursors, transport and biomass burning. *In preparation.*

Presentations and Training

- 7th Joint ECMWF/ESA/EUMETSAT Training on Atmospheric Composition, Royal Belgian Institute for Space Aeronomy, Brussels, 13-17 Oct 2025.
- iCACGP-IGAC 2025 ECR Online Conference, 25 Sep 2025. Ozone and particulate nitrate formation responses to precursors over a South Asian megacity: A multi-satellite approach integrated with ground-based observation.
- Indian Clean Air Summit 2025, Bangalore, 20-22 Aug 2025. Ozone and PM_{2.5} chemical regimes over Delhi: A multi-platform observational study using satellite and ground-based networks.
- IASTA 2024, Doon University, Dehradun, 22-25 Oct 2024. The persistent burden of air pollutants: A long-term trend analysis over India's major metropolises.
- URSI-RCRS 2024, GEHU Bhimtal, 22-25 Oct 2024. Long-term changes in air pollutants at urban regions of India using remote sensing data.
- iCACGP-IGAC Conference 2024, Kuala Lumpur, 9-13 Sep 2024. Decadal variabilities in tropospheric ozone over the Central Himalayas and major urban centers in India.
- National Space Science Symposium 2024, Goa University, 26 Feb 2024. Long-term changes in surface ozone over a high-altitude site in the Central Himalayas.
- TROPMET 2023, IMS Jaipur, 22-24 Nov 2023. Long-term changes in tropospheric ozone over the Central Himalaya: Role of different processes.

- iCACGP-IGAC 2023 ECR Online Conference, 17 Nov 2023. Long-term changes (2007-2022) in surface ozone at a high-altitude site in the Central Himalayas.
- 5th ACAM Workshop, Dhaka, 8-10 Jun 2023. Surface ozone at a high-altitude site in the Central Himalayas: Long-term trends and role of different sources/factors.

Awards

- **CSIR-JRF Award 2020:** Council of Scientific and Industrial Research Junior Research Fellowship; All India Rank 124.
- **CSIR-JRF Award 2019:** Council of Scientific and Industrial Research Junior Research Fellowship; All India Rank 110.

Mentoring, Service, and Outreach

- Mentored three master's and undergraduate students using air pollutant data and satellite remote sensing data for a project on biomass burning impacts on Delhi's existing air pollution during ARIES Training School in Observational Astronomy and Atmospheric Sciences (ATSOAA) 2025.
- Served on the local organizing committee for URSI-RCRS 2024 and the ATSOAA 2024 and 2025.
- Volunteered in science outreach and workshop organization, including the 3rd BINA Workshop, AREA 2023, Aditya-L1 Science Support Cell Workshop 2023, Young Astronomer's Meet 2022, and ARIES National Science Day celebrations.

References

- Dr. Manish Naja (PhD supervisor)
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